

The relationship between academic achievement and off-task social media and smartphone usage: Evidence from a systematic literature review

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ABSTRACT

Systematic cumulative evidence on the effects of the off-task recreational use of social media and smartphones on academic performance among children and adolescents is limited. We conducted a systematic literature review on the impact of off-task usage of smartphones on school achievement among children and adolescents between seven and 18 years, following the Preferred Reporting Item for Systematic Reviews and Meta-analysis guidelines. The literature search was conducted using three electronic databases: Scopus, Web of Science, and Ebscohost. Studies published between January 2013 and June 2025 were included in the search. The systematic review included 34 studies. Results suggested an interrelation between off-task smartphone and social media use and academic achievement, and the relationships detected were negative in most of the studies reviewed.

1. Introduction

Over the past 15 years, interest in the impact of social media and smartphone use on adolescents and young adults has exploded. The primary focus of research has been on the mental health influences of social media and smartphones. The accumulating evidence, although still partly inconsistent, indicates that excessive social media or smartphone use increases the likelihood of suffering from mental health issues such as anxiety and depression (Huang, 2022; Kosola et al., 2024; Marino et al., 2018; Riehm et al., 2019; Sohn et al., 2019).

So far, less attention has been paid to the potential effects of social media or smartphone use on children's school achievement, although academic achievement is an indicator of long-term wellbeing, study trajectory, and further professional success (Paterna et al., 2024). Studies on primary and secondary school students are particularly uncommon. In fact, most existing studies on social media and smartphone use effects on learning outcomes have been conducted among upper secondary school, college or university students (Amez & Baert, 2020; Masalimova et al., 2023; Ojukwu et al., 2021; Otchie & Pedaste, 2020). Some evidence suggests that social media and smartphone use may have

negative effects on children's and adolescents' academic achievement, yet the findings appear to be mixed (Kates et al., 2018). For example, a large-scale study ($N = 10,076$) conducted in Canada, demonstrated that using social media for more than 2 h per day was negatively associated with school connectedness in middle school students and with academic performance in both middle school and high school students after adjusting for age, gender, ethnicity, subjective socioeconomic status and substance use (Sampasa-Kanyinga et al., 2019). However, other studies detected no effects (Ainin et al., 2015; Rashid & Asghar, 2016) between social media usage and academic performance. This might be partly because more additive algorithms were developed in 2015 after the studies were conducted, and variations in the algorithms applied between countries. Few prior reviews on the influence of social media or smartphone use on academic performance have focused on either the use of social media for learning, (i.e. on-task use), or have focused solely on problematic/addictive use. Moreover, these studies have not differentiated between the influences of on- and off-task smartphone or social media usage for learning (see reviews/meta-analysis by Adelantado-Renau et al., 2019; Huang, 2018; Kates et al., 2018; Otchie & Pedaste, 2020; Paterna et al., 2024; Sohn et al., 2019). Accordingly, systematic

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cumulative evidence on the effects of recreational off-task use of social media and smartphones on academic performance among children and adolescents is limited. We aim to address this gap in the literature. To the best of our knowledge, this is the first systematic literature review addressing the interrelation between academic achievement and off-task social media and smartphone usage among children and adolescents aged between seven and 18 years.

1.1. Social media and smartphone usage and academic achievement

A reason for mixed results on the effects of social media or smartphone usage may be that their use is not a single entity and should not be studied as one. This also concerns the partly overlapping distinction between social media and smartphone usage. While social media usage refers to internet-based channels that allow users to interact, selectively self-present users, create and share content in social networks (Carr & Hayes, 2015), smartphones can be used for more diverse purposes, ranging from gaming to shopping. Yet, smartphones are the primary devices for using social media. Accordingly, smartphone use covers a much broader spectrum of activities than use of social media, indicating that the effects of off-task social media and smartphone use on academic achievement may differ. The potential effects of the usage of both social media and smartphones on the students' academic achievement is likely to be dependent on how students use them, and for what purposes. For example, in a meta-analysis, Sung et al. (2016) showed a moderate positive effect of 0.523 for the application of mobile devices to academic achievement, whereas a recent meta-analysis on problematic smartphone usage reported a negative effect on academic achievement (Paterna et al., 2024). From the perspective of school achievement, distinction between off- and on-task usage of social media or smartphones appears to be essential, but very few prior reviews or meta-analyses have considered this in their analysis.

On-task usage refers to using social media or smartphones intentionally for learning, such as seeking knowledge, solving problems, sharing, academic resources, cultivating new skills, using learning applications/games, exchanging opinions and collaborating with others. Conversely, off-task usage, refers to social media or smartphone use that is not intended to cultivate learning or to boost academic achievement (Lau, 2017). Such usage typically includes playing online games, socialising with friends online and entertaining oneself. Drawing from a sample of 348 university students in Hong Kong, Lau (2017) found a non-significant relationship between mobile phone use for academic purposes and student achievement, while a statistically significant negative relationship was found between mobile phone use for non-academic purposes and academic outcomes. While some have found that the more students engage in course-related activities on their phone, i.e. on-task usage, the more likely they are to seek out additional information to comprehend the learning material (Rashid & Asghar, 2016), others report that smartphone usage in the classroom is associated with lower academic achievement (e.g. Amez & Baert, 2020; Ibrahim et al., 2018; Kim et al., 2019). Physical removal of smartphones during class is associated with higher levels of course comprehension (Huey & Giguere, 2023), particularly among low achieving students (UNESCO, 2023). It is important to note that most of these studies have been conducted with college or university students. However, the effects of smartphone and social media recreational use for learning are likely to be dependent on the minors' cognitive and social developmental phase. For example, effects of social media use among adolescents may be highlighted due to the hormonal, physiological, neural, cognitive and social changes puberty changes (Orben & Blakemore, 2023).

Another key characteristic of smartphone or social media use that may have an impact on school achievement is the time spent on off-task use, referring to both the frequency and total time of usage. The time spent on off-task use of social media or smartphones is also one of the criteria of problematic usage (Andreassen et al., 2016). In fact, excessive and addictive use of smartphones is negatively associated with academic

achievement, an increased sense of loneliness, and social anxiety (review by Devi & Dutta, 2022). Problematic smartphone and social media use is not rare. A recent study showed that 16 % of Finnish adolescents aged 15 to 16 years suffered from social media addiction (Kosola et al., 2024). The potential negative causal relationship between excessive social media or smartphone usage and academic achievement may be because less time is available for study activities (Paterna et al., 2024). Also, frequent switching between on-task and off-task activities on smartphones or social media can lead to cognitive overload, inefficiency, lack of attention/concentration, and inability to exert prolonged mental effort, resulting in decreased school achievement (Aru & Rozgonjuk, 2022; Baert et al., 2020; Dontre, 2021). To sum up, prior research and reviews on the effects of smartphone and social media use have focused heavily on a) wellbeing effects, b) made no distinction between the effects of on-task and off-task use, and c) most studies have been conducted on university students and young adults instead of minors. Hence, our systematic literature review adds to the body of knowledge on effects of smartphone and social media use by 1) summarising evidence on their off-task usage for school achievement, and by 2) reviewing the evidence on their use for school achievement among children and adolescents aged between seven and 18 years.

Our aim was to explore the influence of off-task smartphone and social media use on school achievement. The following research questions were addressed in our study: a) how is off-task problematic smartphone use (addiction and nomophobia) and intensity of usage related to school achievement? b) how is off-task problematic social media and internet (addiction) use and intensity of usage related to school achievement? and c) how is off-task purpose of social media or smartphone use related with school achievement?

2. Methods

2.1. Research strategy and data sources

We conducted a systematic literature review based on the guidelines of PRISMA, the Preferred Reporting Item for Systematic Reviews and Meta-analysis statement (Moher et al., 2015). The literature search was conducted using three electronic databases: Scopus, Web of Science and Ebscohost (including the ERIC sub-database) the initial search was conducted in October 2023 and updated in June 2025. Our search exclusively focused on peer-reviewed journal articles, with the deliberate exclusion of various other research document types, including unpublished studies, conference papers, dissertations, literature reviews, meta-studies, editorials, and letters. In instances where the peer-review status was uncertain, verification was conducted for each journal individually. Only articles in English were considered, and the search was limited to the period 2013–6/2025. We chose the past 12 years for our review because in 2013 smartphone penetration rate and social media usage in developed countries reached ca. 50 % (Kakihara, 2014; Poushter, 2016) making internet access and social media usage much easier for children and adolescents, further potentially having an impact on their school achievement. Since then, time spent on smartphones and in multiple social media sites has exponentially grown among minors (Favario & Sidoti, 2024).

We used the following search string: “primary school*” OR “elementary school*” OR “comprehensive school*” OR “secondary school*” OR “high school*” AND (“academic achievement” OR GPA OR “learn* outcome*” OR “learning effect*”) AND (“social media” OR instagram OR tiktok OR snapchat OR youtube OR whatsapp OR “web 2.0” OR “mobile phone*” OR smartphone* OR cellphone*). In all databases, the search was limited to articles published in English, and the publication type was restricted to articles. In Scopus, the search string was applied to the fields *Article Title*, *Abstract*, and *Keywords*. In Web of Science, the *Topic* field was used, which searches the *Title*, *Abstract*, *Indexing*. In EBSCOhost, the search was run separately for the *Title* and *Abstract* fields, which were then combined. The initial search identified

244 studies (114 from Scopus, 77 from Web of Science, and 53 from EBSCOhost). The up-dated literature search added 105 studies (47 from Scopus, 30 from Web of Science, and 28 from EBSCOhost). We consulted the library information specialist when creating the search protocol. Various search string combinations were also pilot tested to optimise the search.

2.2. Inclusion criteria and study selection

We applied the following seven inclusion criteria for the papers to be included:

- Empirical studies assessing the relationship between social media or smartphone use and academic achievement.
- The studies were written in English.
- The studies were published in an international peer-reviewed journal between 2013 and June 2025.
- The participants were students attending general education at a primary or secondary school, with ages ranging between seven and 18 years. Studies focusing exclusively on students with special needs or with diagnosed disorders were excluded.

- The studies addressed students' social media or smartphone usage beyond educational activities, such as teaching, learning, and studying. Accordingly, studies on social media or smartphone usage for educational activities, e.g. use of learning apps, were excluded.
- The studies assessed outcome variables measuring students' academic achievement or academic performance (grade point average (GPA), school subject grade, test scores or self-evaluation).

2.3. Data extraction

Data extraction was conducted manually in three phases. Fig. 1 presents the PRISMA method used to collect and analyse data. First, all the retrieved articles were downloaded to the Zotero reference management program and duplicates ($n = 96$) were automatically identified and removed by the program. After reading the titles and abstracts, 86 articles were included in the full text review. Two authors independently reviewed the remaining articles against the inclusion criteria. Small differing views between researchers were resolved through joint discussion. Thirty-four studies were included in the final analysis and review.

In the last phase of the extraction, relevant information was obtained

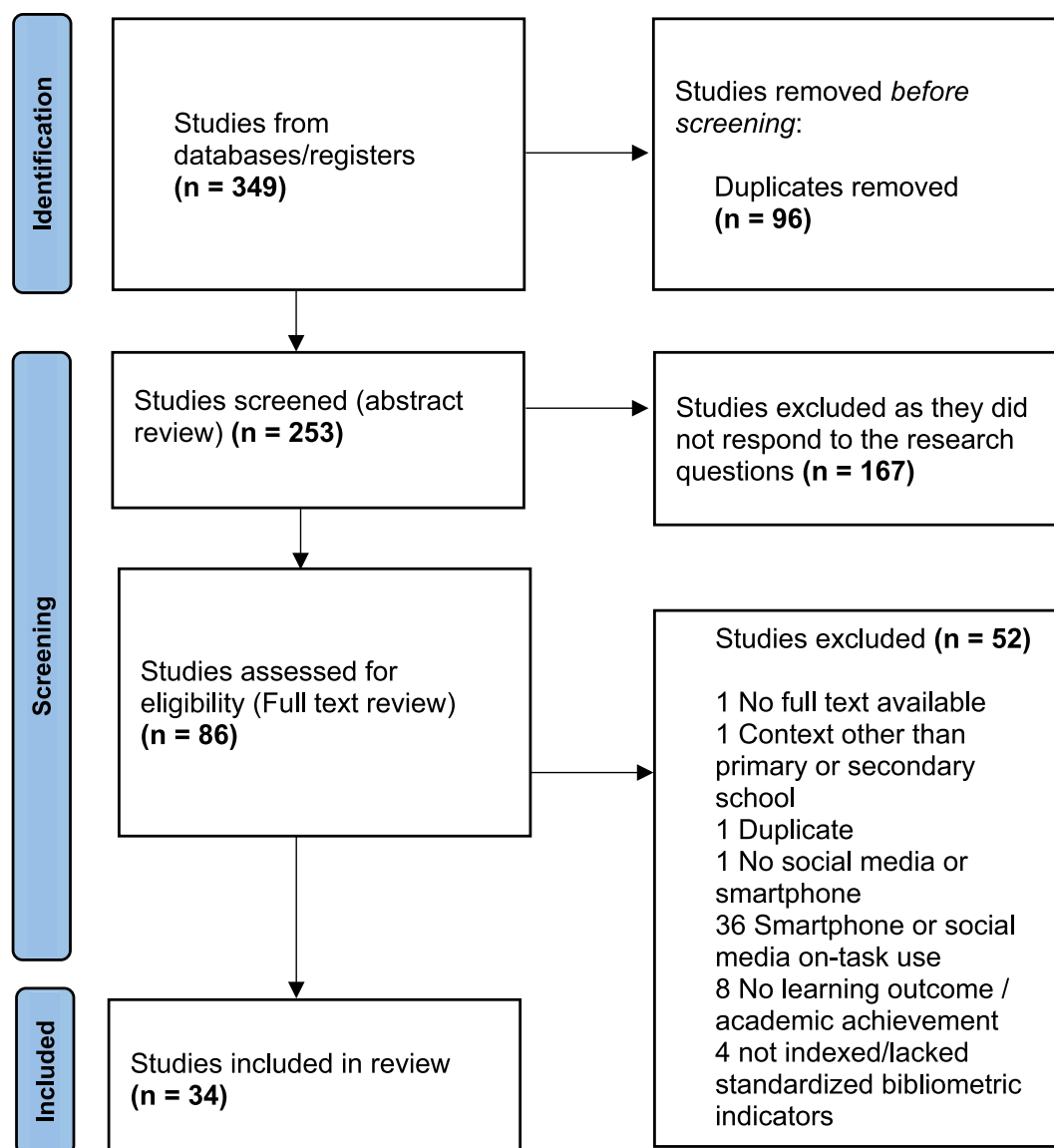


Fig. 1. PRISMA diagram depicting the flow of the systematic review, from the initial identification of 349 database hits to the ultimate inclusion of 34 studies.

from the 34 articles and compiled into a table. The information extracted were year of publication, country in which the study was conducted, school level, number of participants, the role of social media or smartphones in the study, measurements and methods used, learning outcomes and main results. Measured smartphone and social media exposure were categorised as: 1) smartphone addiction, 2) nomophobia, 3) smartphone use intensity, 4) smartphone experience or age at first acquisition, 5) social media and internet addiction, 6) social media and internet use intensity, 7) purpose of social media or smartphone use.

Initially, we searched Scopus, Web of Science, and EBSCOhost. After retrieving bibliometric information and indexing status, we identified four articles that were available only via EBSCOhost. These journals were not indexed in Scopus or Web of Science and lacked standardized bibliometric indicators (Impact factor or CiteScore). To ensure consistency in journal quality assessment and comparability across studies, we excluded these four articles from the final analysis. Consequently, all studies included were sourced from Scopus or Web of Science.

2.4. Quality assessment - Mixed Methods Appraisal Tool (MMAT)

The quality of the articles was assessed using the Mixed Methods Appraisal Tool (MMAT) (Pluye et al., 2011). The MMAT is a critical appraisal tool designed to evaluate the quality of mixed methods research. It assesses various aspects of research components, providing a structured framework to ensure comprehensive evaluation of research quality. Each of the 34 articles was independently evaluated by one of the reviewers, and 14 of the studies were blind rated independently by a second writer. Conflicts were assessed, resulting in a kappa value of 0.84 ($p < 0.001$), indicating 94.3 % agreement (Cohen, 1960). The MMAT protocol has also been used for study classification. All the studies were of non-randomised quantitative design without an intended intervention, and based on the MMAT guidelines, we evaluated: (1) the representativeness of the target population, (2) the appropriateness of the measures used, (3) data completion, (4) use of confounders, and (5) exposure. The studies were assessed as meeting the criteria or not for each component, and each of them represents 20 % of the total quality score (see Appendix 1). When meeting the criteria could not be assessed based on the article in question or immediate reference, we did not contact the authors for further information.

3. Results

3.1. General characteristics of the studies included

We included 34 empirical studies involving 46,093 students in this systematic review. General characteristics of the studies are presented in Table 1. The studies were conducted in 21 countries, most commonly in Europe ($n = 15$) and Asia ($n = 12$) and 21 of them in OECD countries.

The overall volume in the current review's subject area has shown a modest upward trend over time (Fig. 2.). Of the studies reviewed, 20 % (7/34) were published between 2015 and 2019, while the remaining 80 % (27/34) between 2020 and 2025. After the onset of the COVID-19 pandemic, there was a noticeable increase in the number of studies, especially those focusing on smartphone addiction. However, when examining time trends based on the findings of the studies, we did not detect a consistent pattern.

The number of participants per study ranged from 113 to 12,611. Most participants were teenagers. Twelve of the studies were conducted among upper secondary school students (10th–12th grade), seven included secondary school students (7th–9th grade) and ten studies involved students from both school levels. Only one study involved primary school students (1st–6th grade). Three studies included students from both primary and secondary schools, and one study involved students from all three school levels. All studies were of quantitative non-randomised design. Four studies utilised a cohort setting (Dubuc et al., 2020; Gerosa & Gui, 2023; Poulain et al., 2018; Schwartz et al.,

2019) and the rest were cross-sectional analytic studies.

In most of the studies, academic achievement was measured with recorded GPA ($n = 17$), while four of them also reported separately analysed school grades and three self-reported grades. In eight studies, the measure of academic achievement was based on student self-evaluation of their academic achievement or performance. Six studies used standardized or validated skill or performance tests.

Based on the MMAT assessment, only three of the studies fulfilled the criteria for all the components that we evaluated. Most studies received a MMAT score of 80 % ($n = 16$) or 60 % ($n = 14$). Mainly, the expected criteria were not met regarding sample representativeness and use of confounders. Only seven studies were found to have samples that accurately represented the target population, and 19 had included confounders in their analysis (see Appendix 1.).

3.2. The association between academic achievement and smartphone and social media exposure

3.2.1. Smartphone addiction and nomophobia

Most of the studies focused on the association between smartphone addiction and academic achievement ($n = 13$). The Smartphone Addiction Scale–Short Version (SAS-SV) was used in seven studies (Buabbas et al., 2021; Buctot et al., 2021; Gerosa et al., 2022; Guclu et al., 2024; Göger & Kahrman, 2024; Przepiorka et al., 2021; Przepiorka et al., 2025), and two studies used the longer version of the scale (SAS) (Kutluay & Karaca, 2025; Yildiz Durak, 2019). Seven of these studies detected a negative association between the SAS-SV score and academic achievement (Buabbas et al., 2021; Buctot et al., 2021; Göger & Kahrman, 2024; Guclu et al., 2024; Kutluay & Karaca, 2025; Przepiorka et al., 2021; Przepiorka et al., 2025). However, Przepiorka et al. (2025) reported direct negative association between problematic smartphone use and lower GPA only among girls, whereas among boys, no such relationship was detected. Gerosa et al. (2022) reported non-significant associations between SAS-SV and language and mathematics scores, while Yildiz Durak (2019) reported that SAS correlated with higher grades in information technology but found no statistically significant correlations for mathematics, science, English, or social sciences grades.

Other measures used to assess smartphone addiction in the studies reviewed were The Smartphone Application-Based Addiction Scale (SABAS) (Ekşi et al., 2020), Smartphone Pervasiveness Scale for Adolescents (SPS-A) (Gerosa et al., 2022), Mobile Phone Dependency scale (Seo et al., 2016), Mobile-Related Experience Questionnaire (CERM) (Bravo-Sánchez et al., 2021) and Problematic Cellular Phone Use Questionnaire (PCPU-Q) (Wu & Siu, 2020). All these studies reported negative associations between smartphone addiction and academic achievement.

Specifically, Gerosa et al. (2022) reported that the SPS-A scale was significantly associated with lower test scores in mathematics and language, although the SAS-SV scale showed no association. According to Seo et al. (2016), higher mobile phone dependency was directly associated with lower grades in mathematics and indirectly with lower grades in Korean language arts, with attention, depression, and social relationships as mediating factors. Bravo-Sánchez et al. (2021) found that students with low or moderate mobile phone use on the CERM scale had higher grades in Spanish as first language, mathematics and physical education compared to students with high use. Yet, no difference was detected for the grade in English as a foreign language. Göger and Kahrman (2024) showed that adolescents with poor academic achievement had higher odds of suffering from smartphone addiction than their peers with good or very good academic achievement. Similarly, Guclu et al. (2024) reported that internet and smartphone addictions correlated negatively with academic performance.

Three studies assessed nomophobia, i.e. the fear of being without a mobile phone; two of them used the NMP-Q scale (Buctot et al., 2021; Nguyen et al., 2024) while Yildiz Durak (2019) used NPQ -scale to

Table. 1

General characteristics and key results of the included studies.

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
Abu-Shanab and Al-Tarawneh (2015)	Jordan	113	Upper secondary school	Cross-sectional	GPA from four courses	Students' attitudes toward Facebook, user behavior and perceived impact on academic performance (18 items) Time spent on Facebook and internet	Time spent on Facebook and the internet correlated significantly with all dimensions of the measure of Facebook influence. A lower GPA significantly correlated with the dimension indicating excessive time spent on Facebook and the internet.	60 %	International Journal of Web-Based Learning and Teaching Technologies	2.8 (2024, CiteScore)
Bravo-Sánchez et al. (2021)	Spain	501	Secondary and upper secondary school (12–18)	Cross-sectional	Grades in Spanish language and literature, mathematics, English and physical education	Mobile-Related Experience Questionnaire (CERM) 10-item 4-point scale	Students with low or moderate use of mobile phone had higher grades in Spanish, mathematics and physical education compared to students with high use. Significant interactions with mobile-phone-use groups and gender were found, suggesting that girls outperform boys in Spanish and English especially in the group with high mobile phone use.	60 %	International Journal of Environmental Research and Public Health	4.6 (2021, JIF)
Buabbas et al. (2021)	Kuwait	1933	Primary to upper secondary school (11–21)	Cross-sectional	GPA	Smartphone Addiction Scale–Short Version (SAS-SV) 10-item 6-point scale	Students with better school performance were less addicted to smart device use.	80 %	PLoS ONE	2.6 (2024, JIF)
Buctot et al. (2021)	Philippines	3374	Secondary and upper secondary school (13–18)	Cross-sectional	Perceived academic performance (3-point scale)	Smartphone Addiction Scale–Short Version (SAS-SV) 10-item 6-point scale Nomophobia (NMP-Q Scale) 20-item 7-point scale Patterns of smartphone use (incl. Time, frequency, and duration of use, years of experience and purpose of use)	Lower frequency of use on weekdays, purpose of use (i. e. social network sites and social media), higher number of survival days without smartphone and smartphone addiction were significant predictors of low perceived academic performance. Type of internet access (i.e.	80 %	International Journal of Environmental Research and Public Health	4.6 (2021, JIF)

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Table. 1 (continued)

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
							prepaid internet card) and lower number of survival days without smartphone were significant predictors of high perceived academic achievement. Nomophobia was only associated with academic performance in bivariate correlations.			
Caratiquit and Caratiquit (2023)	Philippines	223	Secondary school	Cross-sectional	GPA	Bergen Social Media Addiction Scale (BSMAS) 6-item 5-point scale Academic Procrastination Scale-Short Form (APS-S) 5-item 5-point scale	A significant indirect relationship was detected between social media addiction and academic achievement which was fully mediated by academic procrastination.	60 %	Turkish Online Journal of Distance Education	1.1 (2024, JIF)
Chandrima et al. (2020)	Bangladesh	350	Secondary school and upper secondary (13–17)	Cross-sectional	GPA	Internet Addiction Test Internet use behaviors (incl. Frequency of use, most used apps, and activities) Parental Internet Mediation Scale (PIMS)	Lower GPA, higher weekly and daily use frequency, as well as lower parental mediation were significantly associated with high internet addiction.	80 %	Addictive Behaviors Reports	2.8 (2024, JIF)
Chao et al. (2025)	China	887	Secondary school	Cross-sectional	Self-reported grade, Perceived academic performance	Excessive SNS (Social networking site) use scale, 3-item 7-point scale	Excessive social media use was negatively associated with academic performance.	80 %	Journal of Computer Assisted Learning	4.6 (2024, JIF)
Dhir et al. (2015)	Taiwan/China	417	Upper secondary school (15–18)	Cross-sectional	Grade in the last annual school examination	Compulsive Internet Use Scale (CIUS)	CIUS scores were negatively correlated with the academic performance of the participants.	80 %	Psychiatric Quarterly	2.9 (2024, JIF)
Dubuc et al. (2020)	Canada	199	Secondary school	Cohort study	GPA, final grades	Screen time (television, computer and video games, other computer usage and cellphone) Time spent on social media	At baseline, time spent on social media platforms on weekends and on cellphone on weekdays and weekends was associated with lower GPA among females. No associations were found among males. Longitudinal increase in cell phone use on weekends was associated with decrease in academic	100 %	International Journal of Environmental Research and Public Health	4.6 (2021, JIF)

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Table. 1 (continued)

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
Ekşi et al. (2020)	Turkey	337	Secondary and upper secondary school (14–19)	Cross-sectional	GPA	The Smartphone Application-Based Addiction Scale (SABAS)	performance in mathematics among males during the three-year follow-up. Smartphone addiction levels were significantly higher among students with lower GPA than among students with higher GPA.	60 %	Addicta: The Turkish Journal on Addictions	0.4 (2024, JIF)
Gerosa and Gui (2023)	Italy	1672	Upper secondary school	Cohort study	Standardized language proficiency test score (from INVALSI)	Age at first smartphone acquisition Time spent on screen media (TV and video games)	Variation in language proficiency trajectories was not sufficient to confirm a substantial effect of early smartphone acquisition on language proficiency. Early smartphone acquisition together with heavy baseline screen media use (TV & video games) was associated with decrease in language proficiency trajectory during the three-year follow-up period compared to late owners.	80 %	Social Science Research	3.5 (2024, JIF)
Gerosa et al. (2022)	Italy	3251	Upper secondary school	Cross-sectional	Standardized Italian language and mathematics test scores	Smartphone Pervasiveness Scale for Adolescents (SPS-A) Smartphone Addiction Scale–Short Version (SAS-SV) 10-item 6-point scale	Levels of SPS-A were significantly associated with lower test scores in mathematics and language. Only non-significant associations between SAS-SV and language and math scores.	100 %	Social Science Computer Review	2.7 (2024, JIF)
Göger and Kahrman (2024)	Turkey	546	Secondary and upper secondary school (10–17)	Cross-sectional	Perceived academic achievement (3-point scale)	Smartphone Addiction Scale–Short Version (SAS-SV) 10-item 6-point scale	Poor academic achievement was associated with higher scores of smartphone addiction compared to good or very good academic achievement.	60 %	Addicta: The Turkish Journal on Addictions	0.4 (2024, JIF)
Guclu et al. (2024)	Turkey	1959	Secondary and upper secondary school	Cross-sectional	GPA	Smartphone Addiction Scale, Short Version (SAS-SV), 10 item, 6-point scale	Internet and smartphone addictions were negatively associated with academic performance.	60 %	Revista da Associação Médica Brasileira	1.3 (2024, JIF)

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Table 1 (continued)

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
Kalkan and Cerit (2023)	Turkey	508	Secondary and upper secondary school (12–20)	Cross-sectional	Perceived academic achievement (3-point scale)	Young's Internet Addiction test, short form (YIAT-SF), 12 item 5-point scale Problematic Internet Use Scale-Adolescent (PIUS-A)	Poor academic achievement was found to be associated with problematic internet use.	80 %	Pamukkale Medical Journal	0.2 (2024, CiteScore)
Kokoç (2021)	Turkey	637	Upper secondary school (14–18)	Cross-sectional	GPA	Multitasking with social media (3-item 5-point scale)	Multitasking with social media was significantly associated with poor academic performance.	60 %	Scandinavian Journal of Psychology	2.1 (2024, JIF)
Kutluay and Karaca (2025)	Turkey	410	Upper secondary school	Cross-sectional	GPA	The Smartphone Addiction Scale (SVS) 33 items Time spent using smartphone for social media	Smartphone addiction was negatively associated with academic achievement. The time spent using social media was negatively associated with GPA.	80 %	Education and Information Technologies	5.4 (2024, JIF)
Louragli et al. (2019)	Morocco	541	Secondary and upper secondary school	Cross-sectional	GPA	Bergan Facebook Addiction Scale Test (BFAS) 6-item 5-point scale	Facebook addiction correlated with lower GPA.	60 %	Acta Neuropsychologica	1.1 (2024, JIF)
Morita et al. (2016)	Japan	315	Secondary school (12–13)	Cross-sectional	GPA (from eight school subjects)	Time spent using electronic devices (mobile phone or video games)	Longer time spent using mobile phone or video games correlated with lower GPA, but only among male students.	60 %	Physiology and Behavior	2.5 (2024, JIF)
Mulawarman et al. (2020)	Indonesia	226	Upper secondary school	Cross-sectional	GPA	Questionnaire of social media use motives and frequency	Social media use frequency correlated with lower GPA.	60 %	International Journal of Innovation	0.8 (2024, JIF)
Nguyen et al. (2024)	Vietnam	950	Secondary school	Cross-sectional	Self-reported academic achievement	Nomophobia (NMP-Q Scale) 20-item 7-point scale 20 questions on Smartphone use (inc. access, purpose, frequency, context)	No significant correlation between overall nomophobia and academic achievement. However, some nomophobia components (“Not being able to access information” and “Giving up convenience”) exhibit a significant correlation with GPA among students with moderate nomophobia.	60 %	Computers in Human Behavior Reports	5.8 (2024, JIF)
Oguguo et al. (2020)	Nigeria	150	Upper secondary school	Cross-sectional	Students Accounting Achievement Proforma	Social Media Questionnaire (SMQ)	Study found no association between frequency of use	40 %	International Journal of Evaluation and	2.8 (2024, CiteScore)

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Table. 1 (continued)

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
Poulain et al. (2018)	Germany	850	Secondary school (10–17)	Cohort study	(SAAP), cumulative grade over three terms Grades in mathematics, German and physical education	(frequency of use of social media) Daily time spent using mobile phones	of social media and cumulative grades in accounting. Time spent using a mobile phone at baseline did not predict academic achievement at 12-month follow-up in any subject studied. Total screen time (TV, game console, PC, internet, mobile phone) at baseline predicted achievement in mathematics at follow-up.	80 %	Research in Education BMC Public Health	3.6 (2024, JIF)
Przepiorka et al. (2021)	Poland	427	Primary and secondary school (10–14)	Cross-sectional	GPA	Smartphone Addiction Scale–Short Version (SAS-SV) 10-item 6-point scale	Smartphone addiction correlated with lower GPA.	60 %	Computers and Education	10.5 (2024, JIF)
Przepiorka et al. (2025)	Poland	519	Primary and secondary school (12–16)	Cross-sectional	GPA	Smartphone Addiction Scale–Short Version (SAS-SV) 10-item 6-point scale	In girls, higher problematic smartphone use was directly associated with lower GPA. In boys, problematic smartphone use showed no significant direct relationship with GPA.	80 %	Journal of Early Adolescence	2.2 (2024, JIF)
Robinson et al. (2018)	USA	972	Upper secondary school (17–19)	Cross-sectional	GPA	Years of experience with digital resources at home and school Leisure Computing Index (time spent using social media and entertainment sites) Smartphone Usage Intensity Index (time spent using mobile phone)	Increase in years of experience with digital resources correlated with higher GPA. However, increase in time spent using social media and entertainment on the internet as well as mobile phones correlated with lower GPA.	80 %	American Behavioral Scientist	1.5 (2024, JIF)
Sampasa-Kanyinga et al. (2022)	Canada	4837	Primary and Secondary school	Cross-sectional	Perceived academic performance (5-point scale)	Short Problem Internet Use Test (SPIUT) Time spent using technology (for communication, social media, entertainment, or surfing the internet)	Having symptoms of problem technology use based on the SPIUT scale (score 14 or higher) was associated with lower academic performance. Increase in time	100 %	International Journal of Environmental Research and Public Health	4.614 (2021, JIF)

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Table 1 (continued)

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
Sarafraz et al. (2019)	Iran	315	Upper secondary school (15–18)	Cross-sectional	GPA Educational Performance Test (validated for Iranian society)	Time spent using social media, main networks used and motivation for using	spent using technology (among females over 3 h and among males over 5 h) correlated with lower academic performance. Increased use of social media was associated with lower educational performance score among all participants and with lower GPA among females. At baseline, more frequent use of Instagram, Snapchat and Twitter was associated with poorer academic performance. Frequent use of e-mail correlated with better performance. Longitudinal analyses were not conclusive because academic performance was highly stable across the 12-month study period.	60 %	International Journal of School Health	1.4 (2024, CiteScore)
Schwartz et al. (2019)	USA	376	Upper secondary school	Cohort study	Grades in English, mathematics and science and their GPA	Frequency of using most popular social media (Facebook, Instagram, Twitter, Snapchat), instant messaging systems and e-mail	Higher mobile phone dependency was directly associated with lower grades in mathematics and indirectly with lower grades in Korean language arts, with attention, depression, and social relationships as mediating factors.	80 %	Journal of Youth and Adolescence	3.6 (2024, JIF)
Seo et al. (2016)	South Korea	2159	Secondary and upper secondary school	Cross-sectional	Self-reported grades in Korean language, arts and mathematics	Mobile Phone Dependency 6-item 4-point scale	Problematic mobile phone users (reporting at least four symptoms and at least one functional impairment in PCPU-Q-scale) were more likely to report lower perceived academic performance, more likely to	80 %	Computers in Human Behavior	8.9 (2024, JIF)
Wu and Siu (2020)	Hong Kong	575	Secondary school	Cross-sectional	Perceived academic achievement (5-point scale)	Problematic Cellular Phone Use Questionnaire (PCPU-Q) 12-item binary scale		60 %	Frontiers in Psychology	2.9 (2024, JIF)

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Table. 1 (continued)

Reference	Country	N	School level (age in years)	Study design	Measured academic achievement	Measured smartphone/social media exposure	Key findings	MMAT quality score	Journal	Impact factor
							use mobile phones for social networking, and entertainment, and less likely to use mobile phone for learning or self-management.			
Yamada et al. (2024)	Japan	12,611	Primary school	Cross-sectional	Perceived academic performance (3-point scale)	Screen time (included TV and DVD viewing, video game playing, and Internet browsing only for recreational use)	Academic performance (middle or low vs. high) was not significantly associated with excessive screen time.	80 %	Environmental Health and Preventive Medicine	2.5 (2024, JIF)
Yildiz Durak (2019)	Turkey	612	Secondary and upper secondary school	Cross-sectional	Self-reported grades in mathematics, science, English, social sciences and information technologies	The Smartphone Addiction Scale (SVS) 33 items Nomophobia Scale (NPQ)	Smartphone addiction correlated with higher grades in information technology, but other grades showed no association with smartphone dependency. Nomophobia correlated with lower grades in science.	80 %	Social Science Journal	2.1 (2024, JIF)
Yoon and Yun (2023)	South Korea	2341	Upper secondary school	Cross-sectional	Standardized student scores from mathematics, English, and Korean	Time spent using smartphone for social networking, online lectures, using recreational multimedia, and playing games	Recreational smartphone users had significantly lower academic achievement compared to minimal users or mobile learning users. There was no significant difference between minimal users and mobile learning users.	80 %	Asia Pacific Education Review	2.3 (2024, JIF)

JIF (Journal Impact Factor): Average citations per article over the previous two years, based on Web of Science data.
CiteScore: Average citations per article over the previous four years, based on Scopus data.
Indexed in Web of Science and Scopus: Acta Neuropsychologica, Addicta: The Turkish Journal on Addictions, Addictive Behaviors Reports, American Behavioral Scientist, Asia Pacific Education Review, BMC Public Health, Computers and Education, Computers in Human Behavior, Computers in Human Behavior Reports, Education and Information Technologies, Environmental Health and Preventive Medicine, Frontiers in Psychology, International Journal of Innovation, Journal of Computer Assisted Learning, Journal of Early Adolescence, Journal of Social and Personal Relationships, Journal of Youth and Adolescence, PLoS ONE, Physiology and Behavior, Psychiatric Quarterly, Revista da Associação Médica Brasileira, Scandinavian Journal of Psychology, Social Science Computer Review, Social Science Research, Turkish Online Journal of Distance Education.
Indexed only in Scopus: International Journal of Environmental Research and Public Health, International Journal of Evaluation and Research in Education, International Journal of School Health, International Journal of Web-Based Learning and Teaching Technologies, Pamukkale Medical Journal, Social Science Journal.

measure nomophobia. The results across the studies were relatively inconclusive. Buctot et al. (2021) found a negative connection between nomophobia and self-evaluated academic achievement in bivariate correlations, but no significant associations were found with inferential statistics or multiple regression. Yildiz Durak (2019) reported that nomophobia only correlated significantly with lower grades in science, but not in mathematics, English, social sciences or information technologies. Similarly, Nguyen et al. (2024) found no significant correlation between overall nomophobia and academic achievement whereas some nomophobia components such as “Not being able to access information” and “Giving up convenience” exhibit a significant correlation with GPA

among students with moderate nomophobia.

3.2.2. Smartphone use intensity

Smartphone use intensity was assessed as daily screen time or frequency of use in five studies (Buctot et al., 2021; Dubuc et al., 2020; Morita et al., 2016; Poulain et al., 2018; Robinson et al., 2018). Four of the studies found significant associations between higher screen time or more frequent smartphone use and academic achievement. Two studies, however, reported contradictory results depending on student gender and weekday or weekend use. In one study, screen time was not associated with academic performance (Poulain et al., 2018).

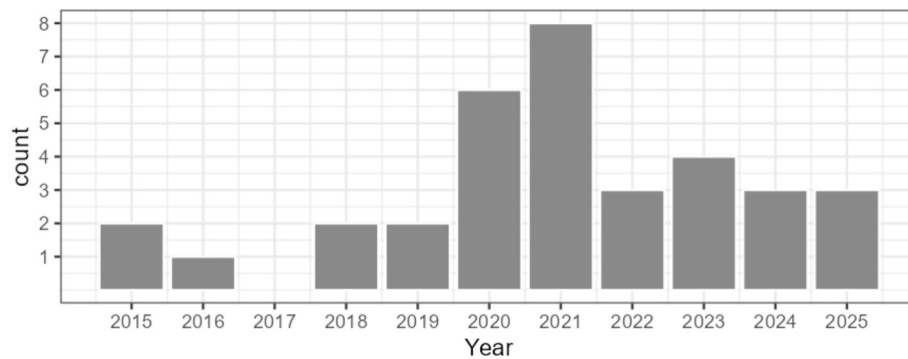


Fig. 2. Number of empirical studies included in this systematic review by publishing year.

According to Robinson et al. (2018), increased time spent using smartphones was significantly associated with lower GPA regardless of gender. However, Dubuc et al. (2020) reported that time spent on smartphones on both weekdays and at weekends was associated with lower GPA only among females. Moreover, a longitudinal increase in smartphone use on weekends was associated with decreased academic performance in mathematics among males. Morita et al. (2016) also detected associations between time spent using mobile phone or video games and lower GPA, but only among male students. Poulain et al. (2018), on the other hand, reported that study time spent using smartphones at baseline did not predict academic achievement at follow-up in any of the subjects studied, but total screen time (TV, game console, computer and smartphone) predicted achievement in mathematics. According to Buctot et al. (2021), lower frequency of smartphone uses on weekdays significantly predicted low perceived academic performance.

Two studies assessed age at first smartphone acquisition or years of user experience (Buctot et al., 2021; Gerosa & Gui, 2023). Buctot et al. (2021) found that a higher number of years of smartphone experience was associated with higher perceived academic achievement. Similarly, Robinson et al. (2018) found that general experience with digital resources correlated with higher GPA. Gerosa and Gui (2023) found no independent association between early smartphone acquisition and language proficiency, but together with heavy screen media use (TV & video games), early acquisition was associated with a decrease in language proficiency trajectory compared to late owners.

3.2.3. Social media and internet addiction

Seven studies specifically assessed social media or internet addiction with various validated measures. The measures were the Bergen Social Media Addiction Scale (BSMAS) (Caratiquit & Caratiquit, 2023), the Internet Addiction Test (Chandrima et al., 2020), the Compulsive Internet Use Scale (CIUS) (Dhir et al., 2015), the Problematic Internet Use Scale-Adolescent (PIUS-A) (Kalkan & Cerit, 2023), the Bergen Facebook Addiction Scale Test (BFAS) (Louragli et al., 2019), the Short Problem Internet Use Test (SPIUT) (Sampasa-Kanyinga et al., 2022), and Young's Internet Addiction test, short version (YIAT-SF) (Guclu et al., 2024). All these studies found significant negative associations between social media or internet addiction and academic achievement. Caratiquit and Caratiquit (2023) found a relationship between social media addiction and academic achievement, which was fully mediated by academic procrastination. According to Chandrima et al. (2020), lower GPA and lower parental mediation were significantly associated with internet addiction.

3.2.4. Social media and internet use intensity

Ten studies assessed the intensity of social media and internet use, most commonly the daily time spent on social media and the internet (Abu-Shanab & Al-Tarawneh, 2015; Dubuc et al., 2020; Kutluay & Karaca, 2025; Mulawarman et al., 2020; Oguguo et al., 2020; Robinson

et al., 2018; Sampasa-Kanyinga et al., 2022; Sarafranz et al., 2019; Yamada et al., 2024). A study on excessive use of social media (Chao et al., 2025). Eight studies reported negative associations between time spent using social media and academic achievement, while only two studies reported no significant associations (Oguguo et al., 2020; Yamada et al., 2024). Like smartphone use, time spent on social media showed a partly divergent relationship with academic achievement between weekends and weekdays, and depending on gender. Dubuc et al. (2020) reported that time spent on social media on weekends was associated with lower GPA only among females. Similarly, Sampasa-Kanyinga et al. (2022) found that an increase in time spent using technology for communication, social media, entertainment, or surfing the internet correlated with lower academic performance. Among females, the negative associations were detected at over three-hours' use, and among males at over five-hours' use. Sarafranz et al. (2019) reported that increased use of social networks was associated with lower educational performance scores (validated in Iran) among all participants and with lower GPA among females.

3.2.5. Purpose of social media or smartphone use

Five studies addressed the time spent using social media or smartphone separately for different purposes. Buctot et al. (2021) found that especially using smartphone for social media was associated with low perceived academic performance. Schwartz et al. (2019) showed that frequent use of popular social networking platforms, such as Instagram, Snapchat and Twitter, was negatively associated with academic achievement. In contrast, frequent use of e-mail correlated with better academic performance. Yoon and Yun (2023) also found that students with profiles as recreational smartphone users had significantly lower academic achievement compared to minimal users or mobile learning users. Kokoç (2021) further showed that multitasking with social media was associated with poor academic performance. Wu and Siu (2020) reported that problematic mobile phone users (reporting at least four symptoms and at least one functional impairment in PCPU-Q-scale) were more likely to use mobile phones for social networking, and entertainment, and less likely to engage in learning or self-management.

4. Discussion

4.1. Limitations of the study

Based on this systematic literature review, research on the effects of non-academic usage (off-task) of smartphones or social media on academic achievement among children and adolescents is still in its infancy. Our search from 2013 to June 2025 resulted in 349 studies, of which 34 studies fulfilled the inclusion criterion. The studies were of quantitative non-randomised design, 30 were cross-sectional, and four were cohort studies. In most studies, academic achievement was measured with GPA and/or grades in certain school subjects that can be considered to be a limited measure of learning at school. Smartphone use was based on self-

reports, and methods for assessing smartphone or social media addiction varied. All studies were correlative or cohort studies, and thus the direction of effect cannot be determined. Hence, the results should be interpreted with caution. Despite its limitations, our review raises concern that off-task smartphone and social media use may interfere with learning, like their effect on adolescent mental health.

4.2. Discussion of key results

Based on the results of our review, the associations between the off-task smartphone and social media use exposure variables and academic achievement variables were mostly negative in the studies we reviewed (Fig. 3). Accordingly, based on the results on using social media and smartphones, non-academic purposes seemed to compromise academic achievement. Especially problematic smartphone and social media usage, including smartphone and social media addiction, were almost without exception negatively associated with academic achievement. Among the studies assessing social media, smartphone, and internet use intensity, the results were far less conclusive. It appeared that smartphone off-task use does not automatically predict poor academic achievement, but rather the effect was dependent on the extent and purposes the device was used for.

Generally, the results regarding social media compared to smartphone use intensity were more pronounced. According to the studies we reviewed, social media use intensity could be more often associated with poor academic achievement than smartphone use per se. Based on the results from the five studies assessing the purpose of use of smartphones, it seems that without exception, the increased use of social media or other entertainment was negatively associated with academic achievement. Moreover, the studies suggested that unhealthy smartphone or social media habits reduce the likelihood of using them for on-task purposes (Wu & Siu, 2020; Yoon & Yun, 2023).

Based on this review, it could not be confirmed that smartphone off-task use intensity negatively affected academic achievement. Even in off-task smartphone use, distinguishing between the specific activities students engage in with their smartphones seems important. Some forms of smartphone and internet off-task use can be beneficial for learning, potentially influencing the results. Yet, in practice, the line between beneficial and problematic off-task smartphone use is fine, since the phone provides access to a variety of apps. Accordingly, straying from on-task usage to off-task is tempting and very easy. Also, constant shifts between the range of smartphone activities are likely to load working memory, and through this, compromise learning goals. To date, very few studies have gathered objective data on smartphone use among adolescents. Among Finnish high school students, data on smartphone use collected from screenshots showed that two thirds of time spent on smartphones was spent on social media, while apps designed for education and creativity were used by a small minority (Kosola et al., 2024). Also in this study, daily time on social media was associated with a lower GPA ($r = 0.280$, $p < 0.001$).

Some of the studies examined the grades of different school subjects separately. Unfortunately, the limited number of these studies hindered drawing definitive conclusions about varying effects across school subjects. However, Yildiz Durak (2019) found that the grade in information technology was positively associated with smartphone addiction. Although Bravo-Sánchez et al. (2021) found negative associations between smartphone addiction and grades in Spanish as a first language, mathematics and physical education, no difference was detected in grades for English as a foreign language. This could indicate that specific school subjects might benefit from frequent smartphone use or might not be as negatively influenced by it.

Some of the studies found significant differences and interactions between genders, but these studies were too few, and the results too inconsistent to draw conclusive conclusions from. Dubuc et al. (2020) found that smartphone and social media use were associated with a lower GPA only among females, but in the longitudinal analysis of the

same study, an increase in smartphone use was associated with decreased academic performance among males. Morita et al. (2016) also found negative associations only among male students. Sampasa-Kanyinga et al. (2022) reported that among females, over 3 h spent using technology was correlated with lower academic performance, while for males, significant associations were detected at over 5 h. The gendered variation may result from the different usage patterns between the male and female students' smartphone use, such as gaming versus social media use (Su et al., 2020).

The studies indicated both direct and indirect mechanisms for the effect of smartphone and social media use on academic achievement. Firstly, time spent on smartphones or social media particularly during weekdays may result in procrastination that lowers the likelihood of success when studying (Caratiquit & Caratiquit, 2023). Secondly, multitasking is likely to break down the long-term concentration necessary for students to learn new things (Kokoç, 2021). Thirdly, decreased wellbeing resulting from problematic smartphone and social media use may further reflect negatively on school achievement. Yet, further evidence on such mechanisms is needed.

The review showed that after the onset of the pandemic, there has been an increase in the number of studies focusing on off-task social media and smartphone usage, more specifically on smartphone addiction. We did not however find a consistent pattern related to the time trends. It is also important to note that the included studies did not address specifically the impact of Covid-19 in terms of off-task social media and smartphone usage and school achievement, the issue should be discussed. Yet, recent meta-studies have shown that both social media and smartphone use, and screen time increased during the pandemic, negatively impacting adolescence wellbeing (e.g. Marciano et al., 2022; Trott et al., 2022) and thus we could expect it to have its toll on academic achievement as well.

Based on the review, several gaps in the literature were identified. Firstly, considering the average time children spend in recreational use of smartphones every day, the body of research on the influence of off-task smartphone and social media use on school achievement is surprisingly scarce. Further studies with a variety of methods on the topic are needed. For example, qualitative studies or mixed methods studies that would allow for a more in-depth understanding of the topic, including examining various platforms, purposes, and forms of smartphone off-task usage were missing. Secondly, since all the studies reviewed were correlative or cohort studies, no conclusions on causal relationships can be made based on our review. Accordingly, to gain evidence on the impact of off-task smartphone and social media use on learning at school, longitudinal and controlled intervention studies are needed. Thirdly, the studies reviewed relied on self-reports in studying smartphone use that are susceptible to various biases and inaccuracies such as memory distortion and social desirability, thus in future studies, using objectively collected data on smartphone use will be required. Fourthly, considering the gendered differences in smartphone off-task use between boys and girls (Abi-Jaoude et al., 2020), it could be assumed that this would result in gendered differences in impact of off-task use on school achievement as well, but so far, the evidence on this is only tentative. Differences might also be based on other student attributes, such as their socio-economic background.

4.3. Practical implications

Due to the correlational nature of the studies, one must be cautious with the implications, but it is still striking that the interrelations detected between off-task smartphone and social media use and school achievement were mainly negative. Firstly, the results indicate that the smartphone and social media use beyond educational contexts and purposes is interrelated with school achievement, and secondly, the relationship is primarily negative. Hence, it appears that off-task use of social media and smartphones may compromise student academic achievement. This implies that more attention should be paid to

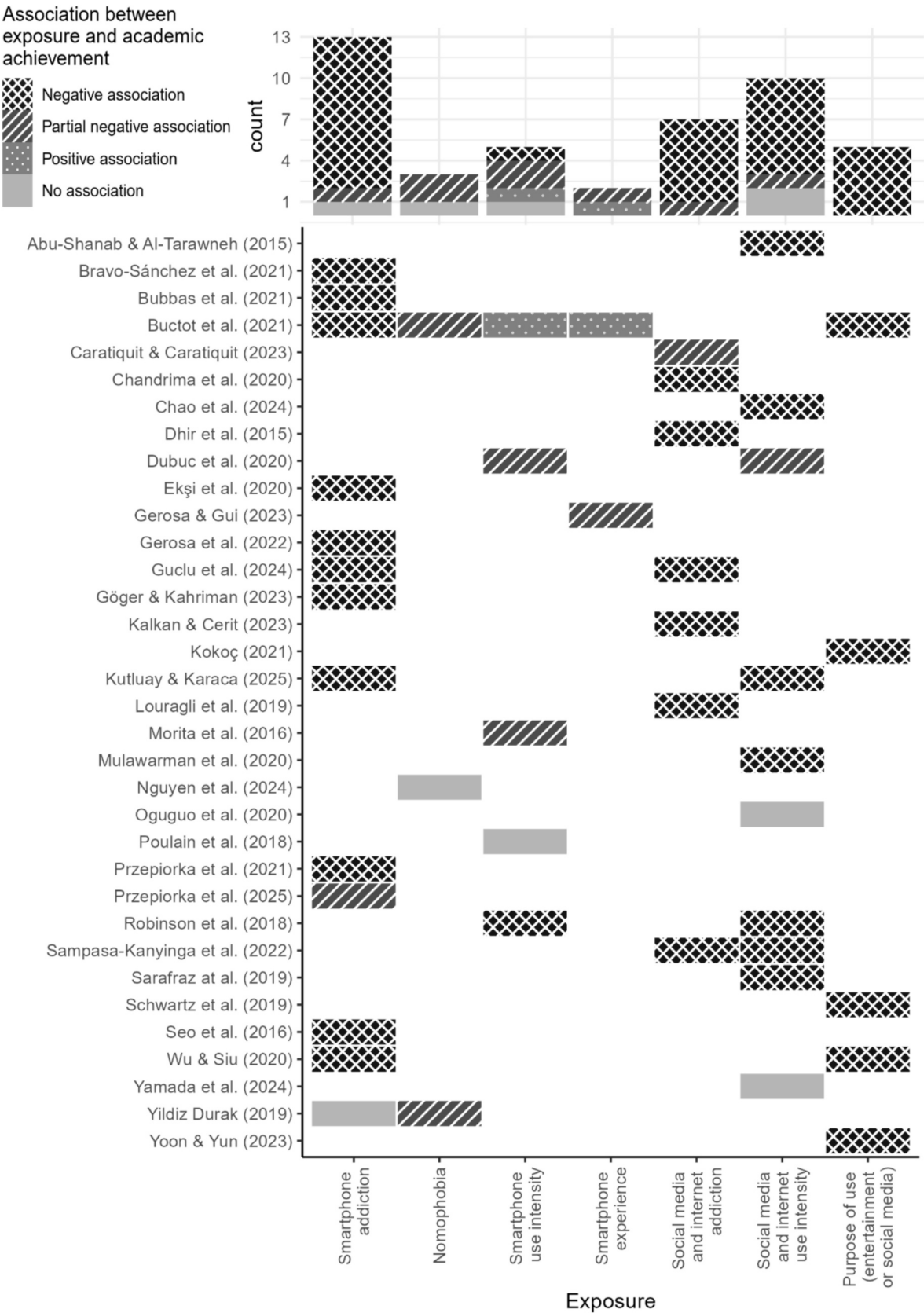


Fig. 3. Heatmap of the general detected associations between smartphone and social media exposure variables and academic achievement variables in the studies reviewed.

children's and adolescents' smartphone and social media use during their free time as well as at school, for example during recess. This is important in terms of both the intensity and purpose of the students' smartphone and social media use, due to the potential negative influence the usage has for the school achievement. In particular, attention should be paid to the forms of use. For example, social media use appeared to be more often negatively associated with the achievement. The results also indicated that unhealthy off-task use may also reflect negatively on on-task use, indicating an interrelation between these two. This suggests that children and adolescents need to be taught how to use smartphones and social media wisely, not only for academic contexts, but also for recreational purposes.

CRedit authorship contribution statement

Kirsi Pyhälä: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology,

Investigation, Funding acquisition, Data curation, Conceptualization. **Hanna Rekola:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis, Data curation. **Salli Salovuori:** Writing – review & editing, Formal analysis, Data curation. **Silja Kosola:** Writing – review & editing. **Henrika Anttila:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation.

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Declaration of competing interest

No potential conflict of interest was reported by the authors.

Appendix 1

Table #. MMAT evaluation of the included articles.

Reference	Study design	Are the participants representative of the target population?	Are measurements appropriate regarding both the outcome and intervention (or exposure)?	Are there complete outcome data?	Are the confounders accounted for in the design and analysis?	During the study period, is the intervention administered (or exposure occurred) as intended?	MMAT score
Abu-Shanab & Al-Tarawneh, 2015	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Bravo-Sánchez et al. (2021)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Buabbas et al. (2021)	Cross-sectional analytic study	Yes	Yes	Yes	No	Yes	80 %
Buctot et al. (2021)	Cross-sectional analytic study	No	Yes	Yes	Yes	Yes	80 %
Caratiquit and Caratiquit (2023)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Chandrima et al. (2020)	Cross-sectional analytic study	No	Yes	Yes	Yes	Yes	80 %
Chao et al. (2025)	Cross-sectional analytic study	No	Yes	Yes	Yes	Yes	80 %
Dhir et al. (2015)	Cross-sectional analytic study	No	Yes	Yes	Yes	Yes	80 %
Dubuc et al. (2020)	Cohort study	Yes	Yes	Yes	Yes	Yes	100 %
Ekşi et al. (2020)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Gerosa and Gui (2023)	Cohort study	No	Yes	Yes	Yes	Yes	80 %
Gerosa et al. (2022)	Cross-sectional analytic study	Yes	Yes	Yes	Yes	Yes	100 %

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Reference	Study design	Are the participants representative of the target population?	Are measurements appropriate regarding both the outcome and intervention (or exposure)?	Are there complete outcome data?	Are the confounders accounted for in the design and analysis?	During the study period, is the intervention administered (or exposure occurred) as intended?	MMAT score
Guclu et al. (2024)	Cross-sectional analytic	No	Yes	Yes	No	Yes	60 %
Göger and Kahrman (2024)	Cross-sectional analytic	No	Yes	No	Yes	Yes	60 %
Kalkan and Cerit (2023)	Cross-sectional analytic study	No	Yes	Yes	Yes	Yes	80 %
Kokoç (2021)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Kutluay and Karaca (2025)	Cross-sectional analytic	No	Yes	Yes	Yes	Yes	80 %
Louragli et al. (2019)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Morita et al. (2016)	Cross-sectional analytic study	No	Yes	No	Yes	Yes	60 %
Mulawarman et al. (2020)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Nguyen et al. (2024)	Cross-sectional analytic	No	Yes	No	Yes	Yes	60 %
Oguguo et al. (2020)	Cross-sectional analytic study	No	No	Yes	No	Yes	40 %
Poulain et al. (2018)	Cohort study	Yes	Yes	No	Yes	Yes	80 %
Przepiorka et al. (2021)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Przepiorka et al. (2025)	Cross-sectional analytic	No	Yes	Yes	Yes	Yes	80 %
Robinson et al. (2018)	Cross-sectional analytic study	No	Yes	Yes	Yes	Yes	80 %
Sampasa-Kanyinga et al. (2022)	Cross-sectional analytic study	Yes	Yes	Yes	Yes	Yes	100 %
Sarafraz et al. (2019)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Schwartz et al. (2019)	Cohort study	No	Yes	Yes	Yes	Yes	80 %
Seo et al. (2016)	Cross-sectional analytic study	Yes	Yes	Yes	No	Yes	80 %
Wu and Siu (2020)	Cross-sectional analytic study	No	Yes	Yes	No	Yes	60 %
Yamada et al. (2024)	Cross-sectional analytic	No	Yes	Yes	Yes	Yes	80 %
Yildiz Durak (2019)	Cross-sectional	No	Yes	Yes	Yes	Yes	80 %

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Reference	Study design	Are the participants representative of the target population?	Are measurements appropriate regarding both the outcome and intervention (or exposure)?	Are there complete outcome data?	Are the confounders accounted for in the design and analysis?	During the study period, is the intervention administered (or exposure occurred) as intended?	MMAT score
Yoon and Yun (2023)	analytic study Cross-sectional analytic study	Yes	Yes	Yes	No	Yes	80 %

Data availability

The articles included in the review are publicly available and presented in Table 1.

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